TheONE AI

Voice-First AI Companion — Architecture & Project Specification

Live App: theone-ai.onrender.com
Repository: github.com/Tarak098/TheONE-AI
Deployment: Render Web Service (FastAPI Full-Stack)

1. EXECUTIVE SUMMARY & DESIGN PHILOSOPHY

TheONE AI is a real-time, voice-first comedic companion built for ultra-low latency, witty, and contextually grounded verbal interaction. Unlike traditional chatbot interfaces with long walls of text, TheONE AI centers interaction around an audio-reactive **Voice Orb**. The user's recognised speech appears on the active screen so the user can verify what the AI heard, while TheONE AI responds **through voice/audio only** with zero live textboxes or response cards cluttering the screen.

2. END-TO-END VOICE PIPELINE

```
[User Mic] ──► 16 kHz Mono WAV Audio
  │
[Gnani Prisma STT v3] ──► Recognized User Speech Transcript
  │ (Displayed immediately in chat area; stored in history)
[Intelligence Engine]
  ├── Unified Natural Comedy Personality (Simple Everyday English)
  ├── Google Gemma 4 26B (with Gemini 2.5 Flash Failover)
  └── Length Validator (Strictly ≤ 1 Sentence, 3–10 Words, Max 15)
  │ (AI response stored in history; voice synthesized)
[Gnani Timbre TTS v2.5] ──► 48 kHz Neural Audio (Yashvi / Zayan)
  │
[Web Audio API] ──► AnalyserNode ──► Reactive Speaking Orb
```

3. NATURAL, SAVAGE COMEDY IDENTITY

The AI sounds like a naturally funny, savage friend who always has an instant comeback ready:

- **Simple Everyday English:** Uses common words spoken in daily conversations. Zero complicated vocabulary, zero textbook jokes, and zero corporate filler.
- **Mood-Independent Humour:** Witty, sarcastic, and bold across all user moods (happy, bored, angry, complaining). Never switches into a dry or boring assistant.
- **Contextual Relevance:** Comedic angle directly tied to the user's specific statement and context. No random unrelated one-liners.

4. STRICT RESPONSE LENGTH LIMIT

Responses strictly adhere to a **maximum of 1 sentence** (preferring 3–10 words, absolute ceiling of 15 words) for punchy comedic timing and rapid voice turnaround.

5. AUDIO-REACTIVE CENTRAL VOICE ORB

The interface is anchored by a dynamic visualizer with 5 states:

- `IDLE` Calm, gentle breathing luminescence.
- `LISTENING` Real-time microphone audio volume reactivity.
- `THINKING` Smooth rotating orbital glow during LLM generation.
- `SPEAKING` Dynamic acoustic ripples reacting to audio playback.
- `ERROR` Gentle warm feedback on network/mic denial.

6. CONVERSATIONAL UI INTERACTION MODEL

- **Recognised User Speech Display:** Prisma STT transcription or typed message appears immediately in the active chat area so the user can see what TheONE AI heard.
- **Voice/Audio-Only AI Responses:** The AI responds strictly via synthesized audio. No AI response textbox, card, or chat bubble is rendered in the active chat area.
- **Full Conversation History Drawer:** Both user messages and AI responses are automatically preserved in the slide-over transcript drawer with voice replay buttons and model tags.
- **Side Scrolling Bar:** Full-page smooth scrolling enabled across all viewports to ensure bottom chat inputs and visualizer remain accessible.

7. NEURAL VOICE SELECTION

Voice	Gender	Engine	Tone & Characteristics
Yashvi	Female	Gnani Timbre (en-IN)	Energetic, sarcastic, upbeat delivery
Zayan	Male	Gnani Timbre (en-IN)	Deadpan, deep, playful comebacks

8. TECHNICAL STACK & PRODUCTION DEPLOYMENT

Backend	FastAPI (Python 3.11), Uvicorn ASGI, Pydantic
Speech-to-Text	Gnani Prisma v3 REST API (16 kHz Mono WAV)
Intelligence	Google Gemma 4 26B with Gemini 2.5 Flash Failover
Text-to-Speech	Gnani Timbre v2.5 REST API (48 kHz WAV)
Frontend	Vanilla ES6+, Canvas 2D, Web Audio API, CSS3
Hosting	Render Web Service (Unified Single-Origin)